ResStock demo, market results

Local energy sharing AMM on measured building data and real day-ahead prices

Community	10 prosumers (5 with rooftop PV, 5 without), single Massachusetts county, ResStock amy2018 release 2
Period	4 trading days, 2018-07-01 to 2018-07-04, 96 sessions of 15 minutes per day
Prices	ISO New England day-ahead LMP, zone WCMA, retail = wholesale + 0.12 EUR/kWh delivery adder
Status quo	every kWh traded with the grid alone: exports paid low (wholesale), imports paid high (retail)
AMM rule	sellers receive r, buyers pay c, both between low and high as in src/Pricing.sol; the residual settles at grid prices

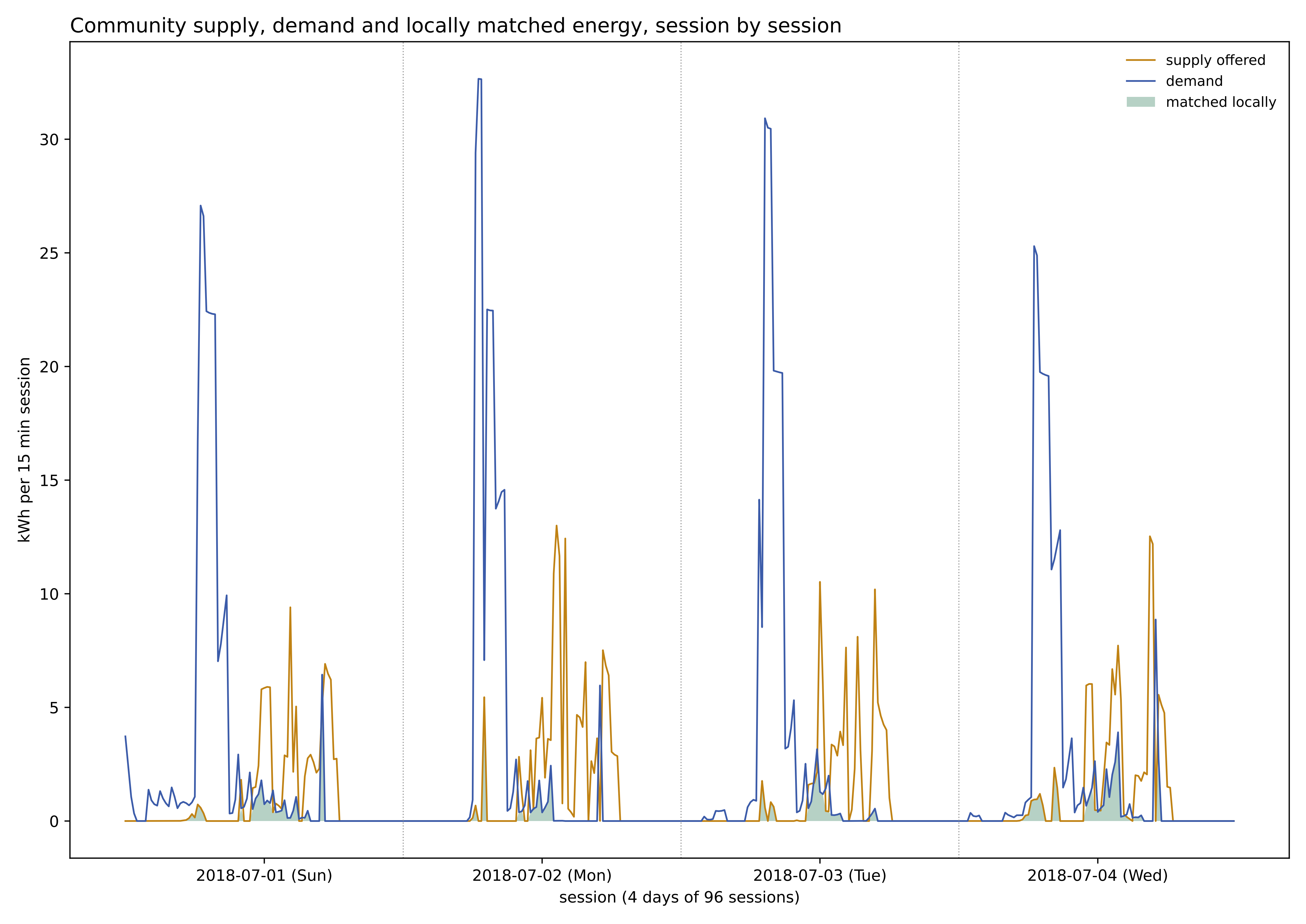
Headline results

- Community gain over 4 days: 9.49 EUR (2.37 EUR/day), entirely the delivery spread earned on locally matched energy
- Locally matched energy: 79.1 kWh of 955.7 kWh demanded, 8.3 % of demand served locally (17.1 % of the surplus absorbed locally)
- Sellers received on average 7.72 c/kWh against 6.69 wholesale; buyers paid 15.00 c/kWh against 15.49 retail
- Spread high - low = the 12 c/kWh delivery adder in every session, so the gain is 0.12 EUR per matched kWh and matching volume is the only lever
- Every member gains or breaks even by construction: r >= low and c <= high in every session

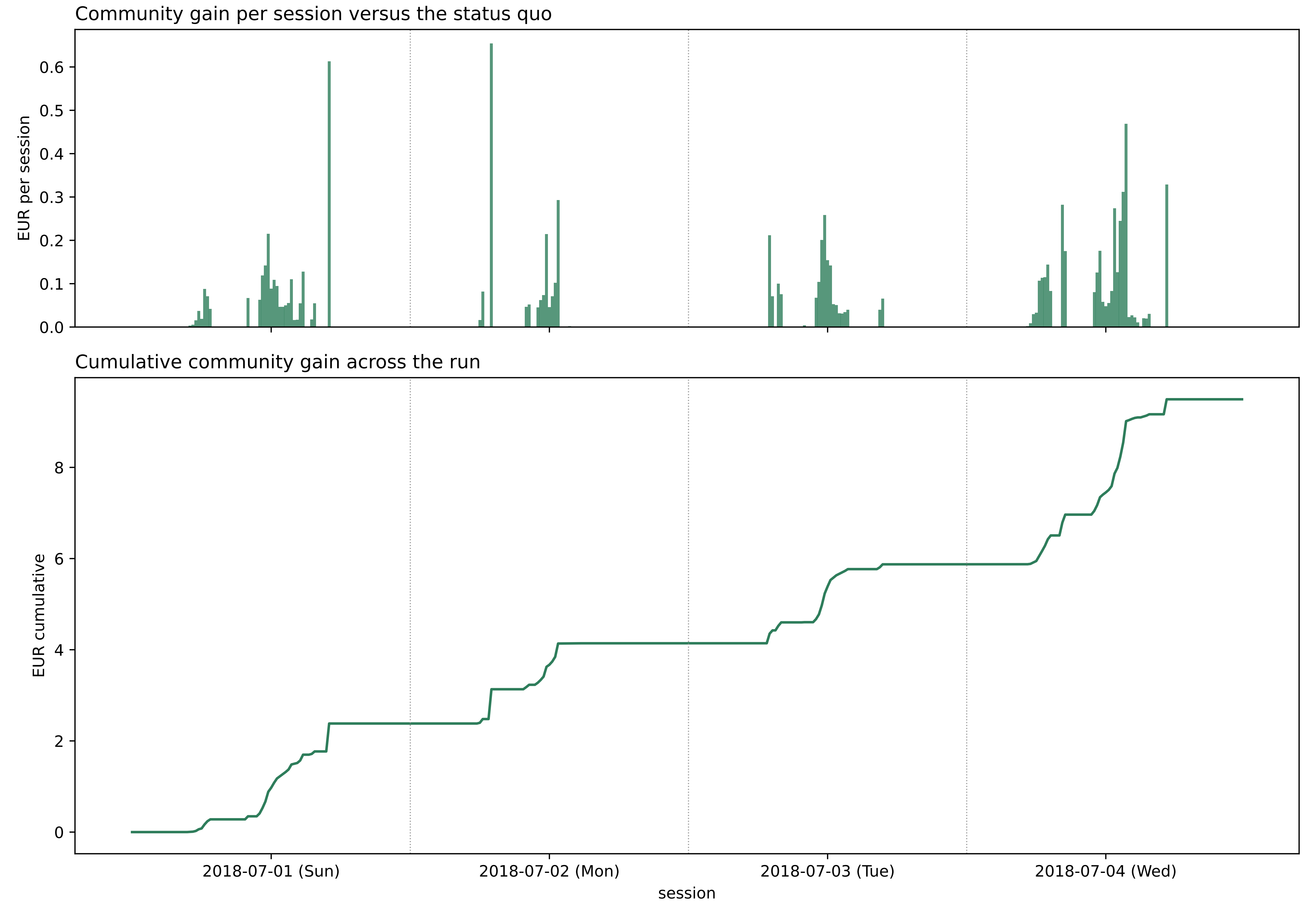
Daily market summary

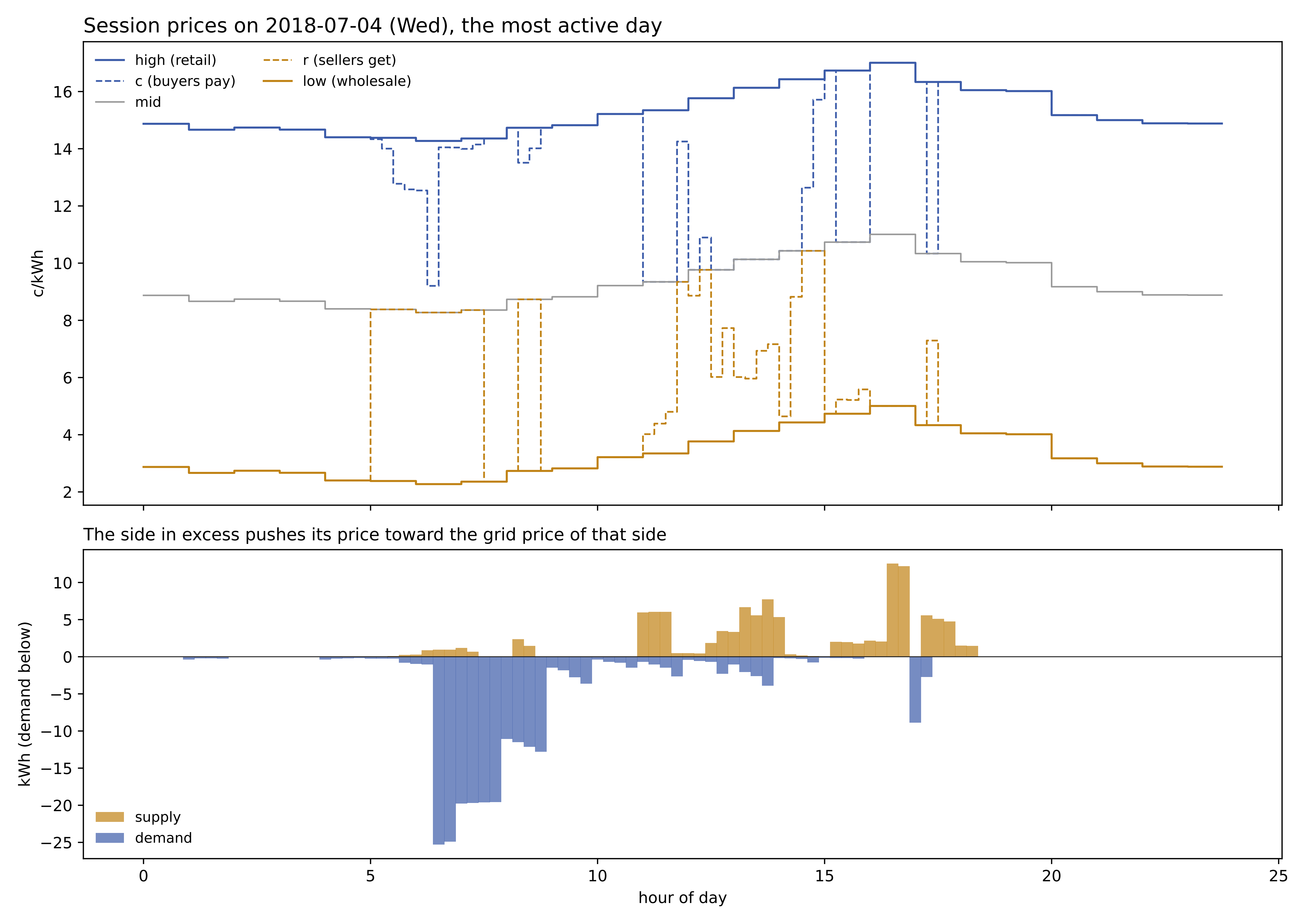
Volumes in kWh. Matched = sum over sessions of min(supply, demand). Covered = matched / demanded, absorbed = matched / offered. The spread high
is the delivery adder, 12 c/kWh in every session, so the gain of a day is exactly the spread times its matched volume.

	offered kWh	demanded kWh	matched kWh	covered %	absorbed %	gain EUR
2018-07-01 (Sun)	102.50	244.88	19.85	8.11	19.36	2.38
2018-07-02 (Mon)	143.91	249.02	14.67	5.89	10.19	1.76
2018-07-03 (Tue)	99.10	233.72	14.43	6.17	14.56	1.73
2018-07-04 (Wed)	116.12	228.10	30.17	13.23	25.98	3.62



Community supply, demand and locally matched energy, session by session
supply offered
demand
matched locally
kWh per 15 min session
0
5
10
15
20
25
30
2018-07-01 (Sun)
2018-07-02 (Mon)
2018-07-03 (Tue)
2018-07-04 (Wed)
session (4 days of 96 sessions)

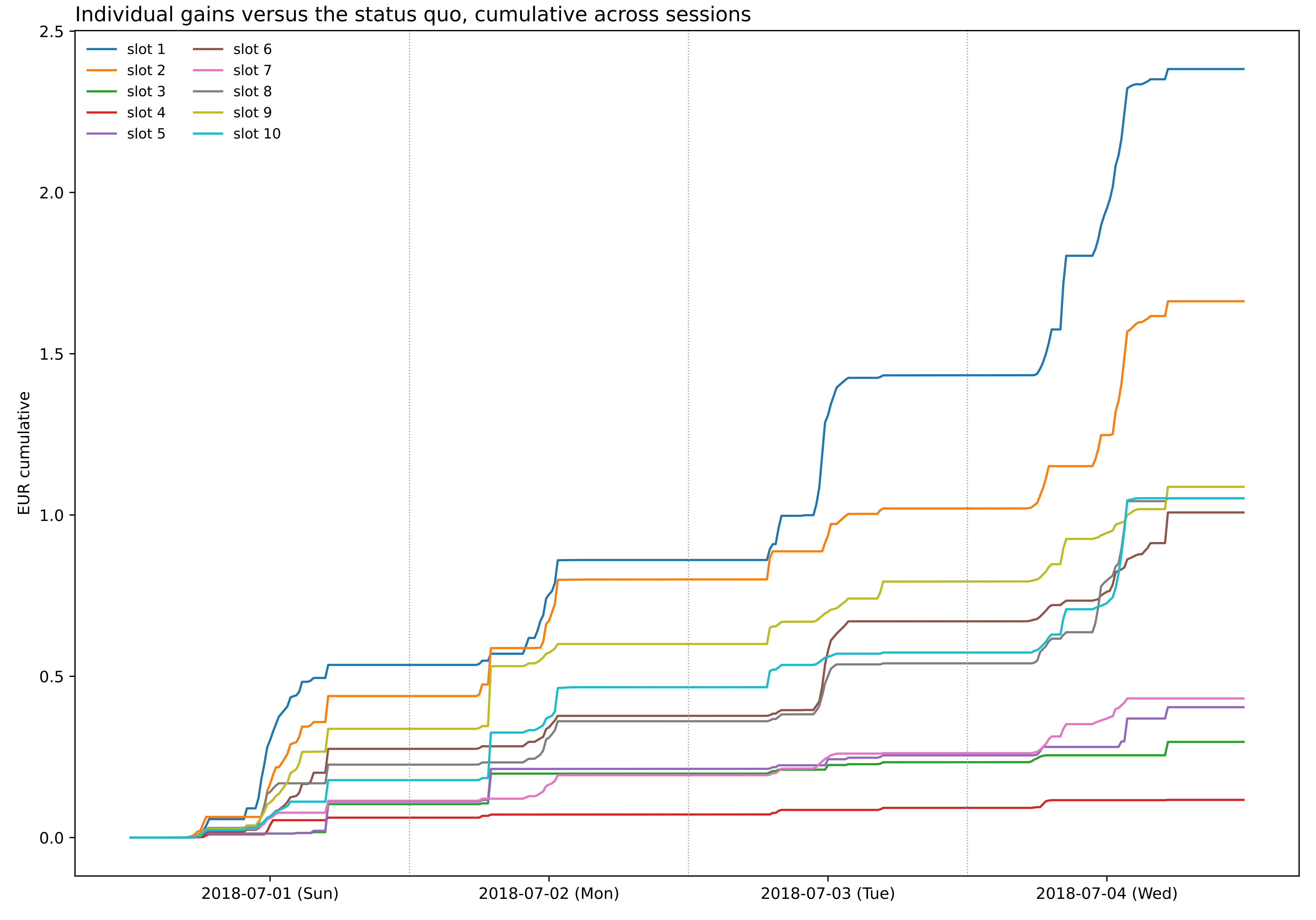

Community gain per session versus the status quo
EUR per session
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Cumulative community gain across the run
EUR cumulative
0
2
4
6
8
2018-07-01 (Sun)
2018-07-02 (Mon)
2018-07-03 (Tue)
2018-07-04 (Wed)
session

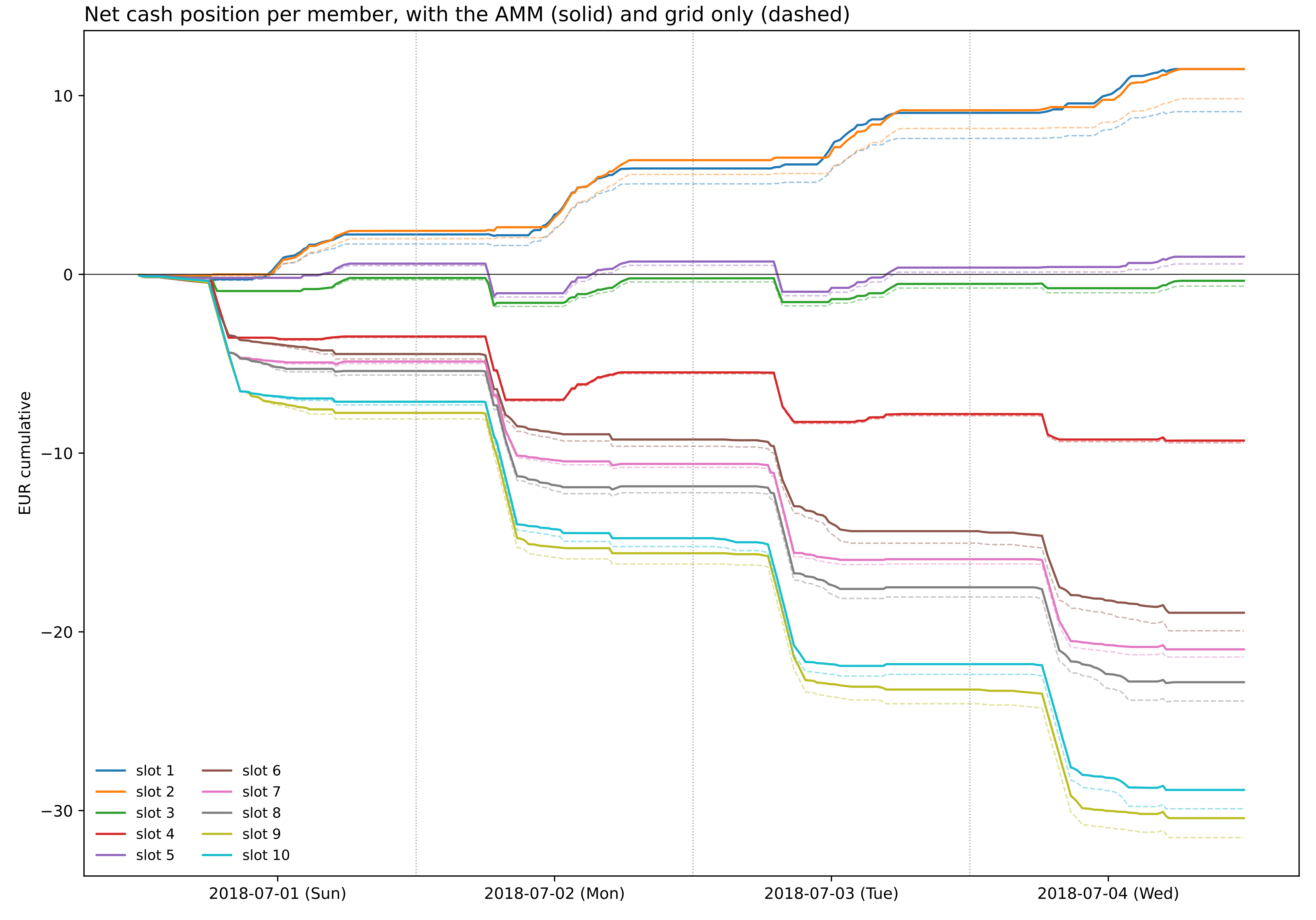

Session prices on 2018-07-04 (Wed), the most active day
high (retail)
c (buyers pay)
mid
r (sellers get)
low (wholesale)
c/kWh
The side in excess pushes its price toward the grid price of that side
supply
demand
kWh (demand below)
hour of day

Individual results over the whole run

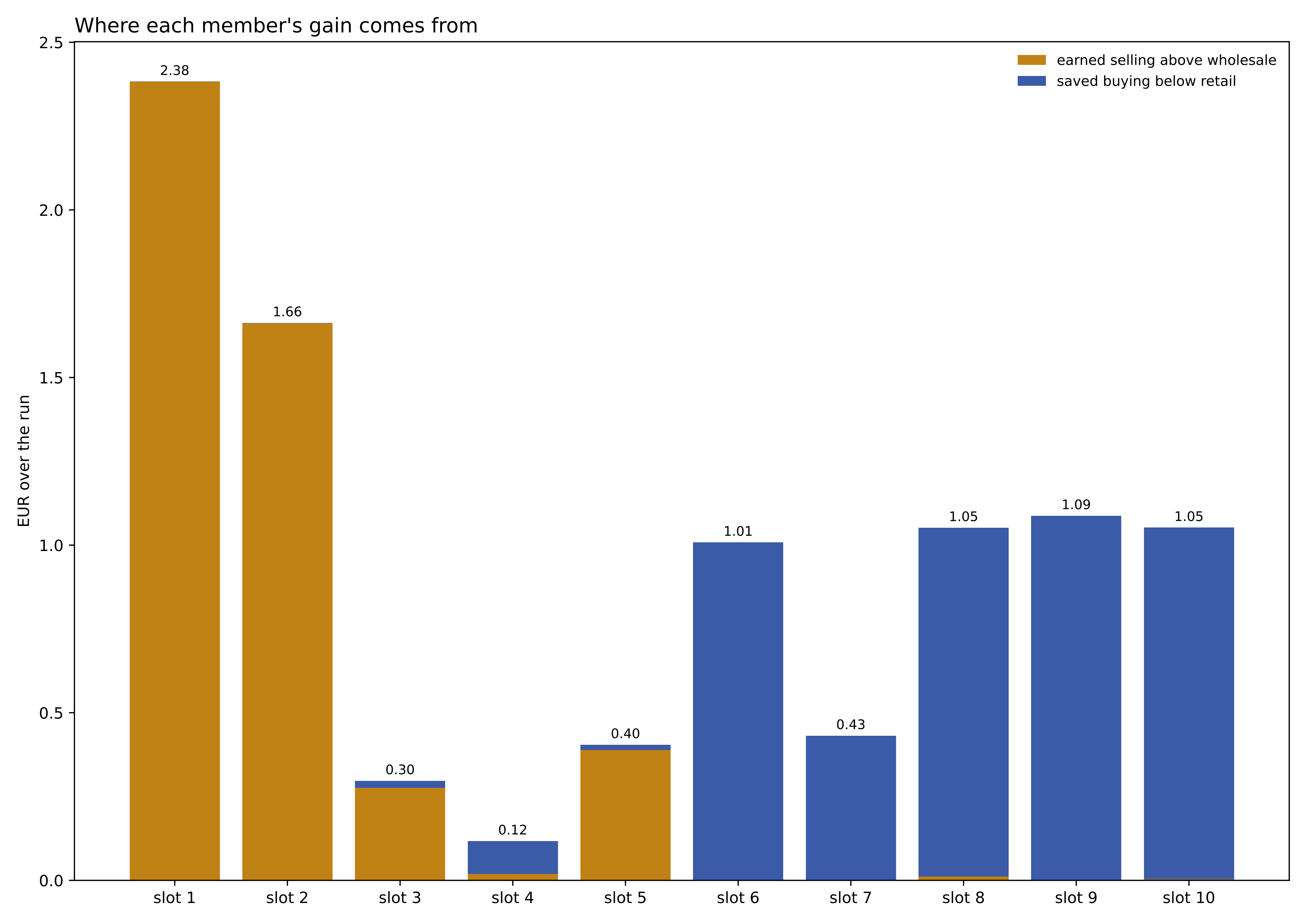
Cash flows in EUR, negative = net payment. Gain = with AMM - grid only; it is non negative for every member in every session.

	sold kWh	bought kWh	grid only EUR	with AMM EUR	gain EUR	gain c/kWh traded	share of gain %
slot 1	153.46	3.69	9.10	11.48	2.38	1.516	25.1
slot 2	153.24	0.91	9.82	11.49	1.66	1.079	17.5
slot 3	48.77	26.86	-0.66	-0.36	0.30	0.392	3.1
slot 4	26.10	76.56	-9.42	-9.30	0.12	0.114	1.2
slot 5	60.19	24.15	0.58	0.99	0.40	0.479	4.3
slot 6	1.82	127.66	-19.94	-18.93	1.01	0.778	10.6
slot 7	5.99	141.55	-21.41	-20.98	0.43	0.292	4.5
slot 8	6.53	156.29	-23.87	-22.82	1.05	0.646	11.1
slot 9	2.24	203.52	-31.51	-30.42	1.09	0.528	11.5
slot 10	3.29	194.52	-29.90	-28.84	1.05	0.532	11.1

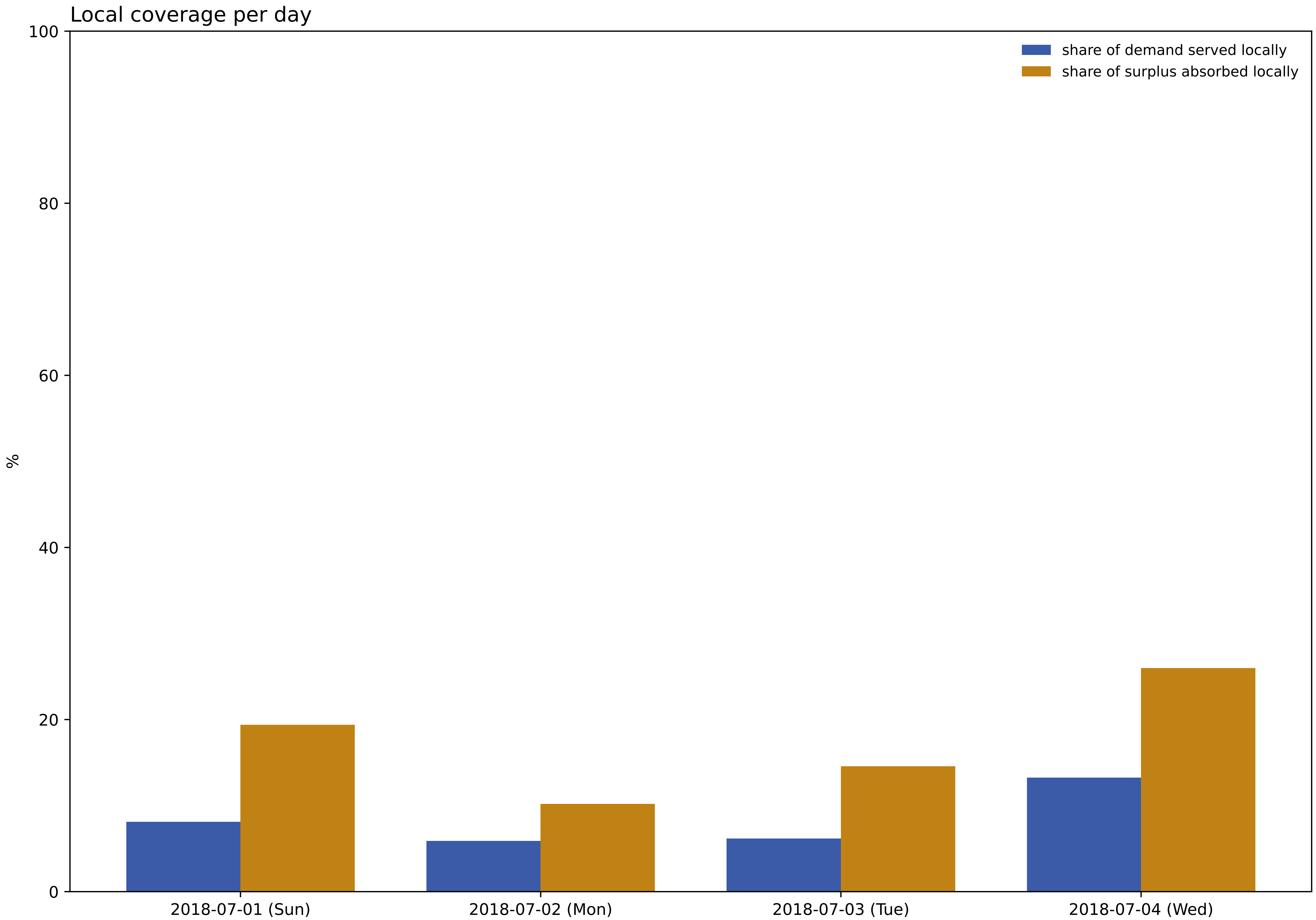

Individual gains versus the status quo, cumulative across sessions
EUR cumulative
2.5
2.0
1.5
1.0
0.5
0.0
slot 1
slot 2
slot 3
slot 4
slot 5
slot 6
slot 7
slot 8
slot 9
slot 10
2018-07-01 (Sun)
2018-07-02 (Mon)
2018-07-03 (Tue)
2018-07-04 (Wed)



Net cash position per member, with the AMM (solid) and grid only (dashed)
EUR cumulative
10
0
−10
−20
−30
2018-07-01 (Sun)
2018-07-02 (Mon)
2018-07-03 (Tue)
2018-07-04 (Wed)
slot 1
slot 2
slot 3
slot 4
slot 5
slot 6
slot 7
slot 8
slot 9
slot 10



Where each member's gain comes from
earned selling above wholesale
saved buying below retail
EUR over the run
2.5
2.0
1.5
1.0
0.5
0.0
2.38
1.66
0.30
0.12
0.40
1.01
0.43
1.05
1.09
1.05
slot 1
slot 2
slot 3
slot 4
slot 5
slot 6
slot 7
slot 8
slot 9
slot 10


Local coverage per day
share of demand served locally
share of surplus absorbed locally
%
100
80
60
40
20
0
2018-07-01 (Sun)
2018-07-02 (Mon)
2018-07-03 (Tue)
2018-07-04 (Wed)

Method and caveats

- Inputs: demo/data/netputs_resstock.json (per member, per day, 96 sessions of [sell, buy] in Wh, the solver's output) and demo/data/prices.json (ISO-NE day-ahead low and retail high in EUR/kWh); paths resolved relative to this script, overridable with --netputs and --prices.
- Session prices reproduce src/Pricing.sol in floating point. The contract rounds r down and c up at a scale of about 1e-7 c/kWh, negligible here.
- Baseline: the same physical quantities traded with the grid alone. The comparison isolates the settlement rule; behaviour is held fixed, so any demand response the tariff itself would trigger without a market is not counted.
- The community gain equals the tariff spread times the matched volume, verified numerically at machine precision in this script. Under net metering (export credited at retail) the spread is zero and the market has no value to distribute.
- Floors, deposits and withdrawal queues are ignored: with the reference DEPOSIT_EUR=200 no member was ever excluded by its floor during the run.
- Gains are split into a seller side (r - low on energy sold) and a buyer side (high - c on energy bought). Both are non negative in every session, so no member is ever worse off than the status quo.
- The panel is a composition choice, not a representative sample: 5 dwellings with PV and 5 without, drawn from one county so the community plausibly shares a feeder and a pricing zone.
- Coverage is measured at 15 minute simultaneity, which is the binding constraint: daily energy volumes overlap far more than sessions do. The solver moves flexible load and battery charging to the cheap night hours, away from the midday PV, so the tariff that creates the spread also desynchronises demand from local supply.
- This report uses the pairing the solver optimised against: load day d with price day d. In the on-chain demo, post_prices.ts cycles the 5 dates of prices.json anchored at the deployment day while the operator cycles the 4 netput days by absolute day number, so the load/price pairing rotates across the run and client balances differ from this report on most days.